A Reliability-Centered Workflow Layer for Tuberculosis CAD in Hospital Chest X-Ray Screening

ALIVE / Prism Draft

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1 Working Thesis

Baseline TB/CXR model feasibility is largely established. WHO has recommended CAD for TB screening since 2021 and, in its June 2025 policy update, reported that six CAD products met WHO performance standards for TB screening in people aged 15 years and older. More broadly, chest X-ray AI reviews consistently report strong diagnostic performance across tasks, while a 2024 study found that AI assistance raised physician AUC from 0.773 to 0.874 and helped non-radiologists perform on par with radiologists. In a real-world reporting study, radiologists showed complete agreement with model findings in 86.5% of reviewed cases, with only 0.5% of findings deemed missed by the model. Taken together, these results justify treating raw model accuracy and basic feasibility as comparatively solved background for this paper, while shifting attention to the unsolved question of reliable hospital integration.[1][2][3][4][5]

This paper should be framed as an *academic prototype and workflow implementation* contribution rather than as "another tuberculosis classifier." The central claim is that chest X-ray deep learning for TB is already strong enough to justify a shift in emphasis toward **prototype feature integration for real hospital use**, with **clinically grounded multimodal explainability** positioned as the lead contribution and **input quality gating, explicit uncertainty-aware deferral, and post-deployment monitoring** as supporting workflow layers. The proposed system therefore combines these features on top of a mature TB scoring backbone. Systematic and implementation-focused reviews repeatedly note that the value of chest X-ray AI in practice depends not only on accuracy, but also on workflow fit, clinician interaction, local validation, and ongoing quality assurance.

Why this narrative is cohesive:

[1] WHO policy statement overview, `https://www.who.int/publications/i/item/9789240110373`.

[2] WHO approval update, `https://www.who.int/news/item/11-06-2025-who-approves-six-software-products-for-computer-a`

[3] Systematic review, `https://pmc.ncbi.nlm.nih.gov/articles/PMC9955112/`.

[4] Pamela G. Anderson et al., "Deep learning improves physician accuracy in the comprehensive detection of abnormalities on chest X-rays," `https://www.nature.com/articles/s41598-024-76608-2`.

[5] Real-world observational study, `https://pubmed.ncbi.nlm.nih.gov/34930738/`.

- It uses model performance only as background motivation, not as the claimed novelty.
- It unifies the selected features under one layer: **reliability**. Multimodal explanation is the lead user-facing feature, while quality gating protects the input, explicit deferral protects the decision, and monitoring protects deployment over time.
- It fits the practical medical deployment angle better than a purely technical CV paper, especially for hospital integration and L1/L2 settings.

Core evidence to anchor the paper:

- Reviews of chest X-ray AI deployment note that only a limited share of studies are prospective or clinically embedded, and that workflow integration, local testing, and human factors remain under-studied.[6]
- Reviews of AI for chest radiographs argue that expertise, usability, and implementation context remain essential even when model performance is strong.[7]
- A systematic review of AI for chest X-ray interpretation highlights strong diagnostic performance overall, but limited prospective validation and real-world evidence.[8]

2 Alternative Narratives Considered

Narrative	Why It Was Attractive	Main Risk / Why Not Selected	Evidence to Cite
TB-specific classifier improvement	Clear CV story; easy to describe as a model paper	Weak novelty because TB chest X-ray CAD is already mature, commercially deployed, and extensively benchmarked; difficult to outperform convincingly within one month	WHO approval of six CAD products for TB on chest X-ray; broad literature on strong CXR DL performance.[9]

[6] M. H. Mongan et al., "A review of practical AI in chest radiography," https://pmc.ncbi.nlm.nih.gov/articles/PMC10128969/.

[7] N. A. M. et al., "Artificial intelligence for chest radiograph interpretation in clinical practice," https://www.mdpi.com/2075-4418/13/4/743.

[8] A. D. et al., "Artificial intelligence for chest x-ray interpretation: a systematic review," https://pmc.ncbi.nlm.nih.gov/articles/PMC9955112/.

[9] WHO, "WHO approves six software products for computer-aided detection of TB on chest X-ray," https://www.who.int/news/item/11-06-2025-who-approves-six-software-products-for-computer-aided-detection-of-tb-on-chest-x-ray

Threshold calibration as main contribution	Strong Philippine relevance; ties to local context and hospital constraints	Already partially addressed by WHO calibration guidance, CAD4TB dynamic-threshold tooling, and a 2025 Philippine qXR study showing the trade-off; good motivation, but not enough as headline novelty	WHO local calibration guidance; Philippine qXR study; CAD4TB product positioning.[10][11][12]
Educational mode as primary narrative	Fits ALIVE's broader mission; helpful for junior clinicians and students	Best kept as a secondary contribution because evidence is less direct and conference fit is stronger when education is attached to a deployment problem	Use implementation and usability literature, plus saliency/interpretability limitations, as indirect support.[13]
Cost-effectiveness / cost-savings paper	High policy relevance; strong practical appeal in the Philippines	Difficult to argue convincingly in this paper because publicly available Philippine cost figures are fragmented across facility tariffs and modeled studies, and they do not directly capture hospital-level AI-CAD workflow costs, downstream utilization, or implementation effects without a local pilot	Can be mentioned in Discussion as future work rather than central claim.

[10] WHO Kobe Centre, "Determining the local calibration of CAD thresholds and other parameters," https://wkc.who.int/resources/publications/i/item/determining-the-local-calibration-of-computer-assisted-detection-(cad)-thresholds-and-other-parameters.

[11] A. Marquez et al., "Performance of chest x-ray with computer-aided detection as a triage test for pulmonary TB in the Philippines," https://bmcglobalpublichealth.biomedcentral.com/articles/10.1186/s44263-025-00198-y.

[12] Delft Imaging, CAD4TB product page, https://delft.care/cad4tb-solution/.

[13] Oakden-Rayner et al., usability study, https://pubmed.ncbi.nlm.nih.gov/36189838/.

Analog-film / smartphone capture for low-resource provinces	Strong L1/L2 relevance; could expand access where digital workflows are incomplete	Domain shift, regulatory scope, and image-quality variability make it risky without dedicated evaluation; safer as a quality-gating or deferral subcase than as the main narrative	Studies on photographed radiographs plus vendor input constraints.[14][15]

3 Product Comparison

In particular, we found the clearest market gap for **clinically grounded multimodal explanation of findings in natural language** and for a **combined user-facing workflow layer** that joins input QA, explicit deferral, and explanation. We found partial evidence of input-quality handling for Siemens through an internal input quality gate, but not comparable clear user-facing evidence in the other reviewed official materials.

This gap requires careful qualification. Some reviewed products already provide textual summaries, structured finding labels, or draft reporting support. However, based on the official materials reviewed for this outline, we did not find clear evidence of a commercially deployed feature that explains a diagnostic conclusion in clinician-facing causal language of the form "TB suspected because of [finding] in [location]." The strongest prior evidence comes from Siemens research, which explicitly studies converting detected abnormalities and probabilities into report text with a fine-tuned language model, but this appears in research rather than as a clearly shipped AI-Rad Companion feature. This matters because the same direction appears to be highly desired by users: usability testing of a chest X-ray AI tool reported requests for clearer explanatory support and broader deployment in understaffed radiology departments, trainee settings, and non-radiologist specialties. The contribution in this paper is therefore framed as *prototype implementation*, not clinical validation, of that still-emerging explanation layer.[16][17]

[14] M. Nasser et al., photographed analog chest radiograph pilot, `https://pmc.ncbi.nlm.nih.gov/articles/PMC12955127/`.

[15] qXR documentation, accepted input workflow, `https://documentation.qure.ai/api/platform-api/get-started/qxr`.

[16] Siemens research on LLM-supported chest X-ray reporting: `https://arxiv.org/html/2306.10448`.

[17] Cheung et al., "U"AI" Testing: User Interface and Usability Testing of a Chest X-ray AI Tool in a Simulated Real-World Workflow," `https://journals.sagepub.com/doi/10.1177/08465371221131200`.

Feature	qXR	CAD4TB+	Lunit INSIGHT CXR	Annalise Enterprise CXR	Siemens AI-Rad Companion	Proposed system

Feature	qXR	CAD4TB+	Lunit INSIGHT CXR	Annalise Enterprise CXR	Siemens AI-Rad Companion	Proposed system

TB support	✓[18]	✓[19]	✓[20]			✓[21]
Localization / overlay	✓[22]	✓[23]	✓[24]	✓[25]	✓[26]	✓[27]
Worklist / prioritization	✓[28]		✓[29]	✓[30]		
Threshold tuning		✓[31]				✓[32]
Monitoring / analytics	✓[33]	✓[34]	✓[35]			✓[36]
Input QA gate					✓[37]	✓[38]

[18]qXR supports chest X-ray interpretation and workflow integration for TB-related screening use cases: https://documentation.qure.ai/api/platform-api/get-started/qxr.

[19]CAD4TB is specifically positioned for TB detection on chest X-ray: https://delft.care/cad4tb/.

[20]Lunit states that INSIGHT CXR supports tuberculosis screening: https://www.lunit.io/en/products/cxr?trk=test.

[21]This outline assumes a mature TB scoring module as the predictive baseline rather than the main novelty of the paper.

[22]Qure documentation describes heatmap thumbnails and overlay highlighting of abnormal areas: https://documentation.qure.ai/release-notes/qure.ai-app; https://documentation.qure.ai/.

[23]CAD4TB provides a radiological heatmap / coloured overlay on the original chest X-ray: https://delft.care/cad4tb/.

[24]Lunit says the system highlights thoracic abnormalities and supports current-prior comparison with progression tracking: https://www.lunit.io/en/products/cxr?trk=test.

[25]Annalise describes coloured overlays on the image to localize findings: https://www.annalise.ai/CXR.

[26]Siemens documentation describes highlighted findings with localization information on the image: https://academy.siemens-healthineers.com/_/en-us/chest-x-ray-confirmation-workflow/; https://academy.siemens-healthineers.com/_/en-us/whitepaper-operating-characteristics-and-algorithm-performance-ai-rad-companion-chest-x-ray/.

[27]Proposed structured evidence interface with localized visual evidence in this paper outline.

[28]Qure provides a worklist panel, AI-based prioritization, notification settings, and critical/routine triage status: https://documentation.qure.ai/users-manual/qure.ai-app-users-manual/qure.ai-app-features/windows-mac-os-app-ui/worklist-panel; https://documentation.qure.ai/users-manual/qure.ai-app-users-manual/qure.ai-app-features/windows-mac-os-app-ui/navigation-toolbar/settings/notification-settings; https://documentation.qure.ai/api/platform-api/api-specifications/fetching-results.

[29]Lunit describes AI-integrated worklists, surfacing critical cases, and workload reduction: https://www.lunit.io/en/products/cxr?trk=test.

[30]Annalise describes AI-powered worklist prioritisation and notification of urgent cases: https://annalise.ai/annalise-enterprise-cxr/; https://www.annalise.ai/CXR.

[31]CAD4TB advertises a dynamic-threshold feature that suggests optimized thresholds based on previous results and testing capacity: https://delft.care/cad4tb/.

[32]Thresholding is retained as a configurable part of reliability logic, not as the paper's main novelty.

[33]Qure markets qTrack/customized dashboards to monitor program health and care coordination: https://www.qure.ai/.

[34]CAD4TB describes an insights module for management and centralized synchronization across sites: https://delft.care/cad4tb/.

[35]Lunit markets INSIGHT Manager 2.0 as integrated monitoring and analytics for day-to-day use and setting refinement: https://www.lunit.io/en/products/cxr?trk=test.

[36]Proposed post-deployment monitoring module in this paper outline.

[37]Siemens describes a pre-defined DICOM input quality gate; for this comparison, input QA is therefore treated as subsumed by their documented input quality gate rather than exposed as a separate user-facing workflow layer: https://academy.siemens-healthineers.com/_/en-us/whitepaper-operating-characteristics-and-algorithm-performance-ai-rad-companion-chest-x-ray/.

[38]Proposed input QA gate in this paper outline.

Feature	qXR	CAD4TB+	Lunit INSIGHT CXR	Annalise Enterprise CXR	Siemens AI-Rad Companion	Proposed system
Explicit uncertainty / deferral					✓[39]	✓[40]
Natural-language multimodal explanation						✓[41]

4 Selected Narrative: A Reliability-Centered Workflow Layer for TB CAD Deployment

4.1 One-Sentence Claim

We propose a **prototype reliability-centered workflow layer** for tuberculosis CAD on chest X-ray that improves hospital deployability by adding a clinician-facing multimodal explanation layer for findings in natural language, while also screening poor-quality inputs and explicitly deferring uncertain cases, rather than claiming novelty in the underlying TB classifier itself. Post-deployment monitoring is retained as a supporting operational layer.

4.2 Reliability Layer Modules

Mature baseline module:

- **TB scoring module (mature / not the main novelty).** Use a practical TB classifier based on prior literature and public datasets as the predictive core, while making clear that this module is inherited from a mature research line rather than presented as the paper's main novelty.[42][43]

New workflow-layer modules proposed in this paper:

1. **Clinically grounded multimodal explainability module.** Add a VQA / natural-language explanation layer that translates model findings into user-facing textual explanations of suspected abnormalities and their visual basis. This is the lead prototype feature in the paper.

[39] Siemens presents AI-confidence scores and thresholded highlighting, but not an explicit deferral workflow: `https://academy.siemens-healthineers.com/_/en-us/chest-x-ray-confirmation-workflow/`.

[40] Proposed uncertainty-aware deferral module in this paper outline.

[41] Proposed clinically grounded multimodal explainability layer: a VQA / natural-language explanation module for findings in this academic prototype. Official materials reviewed for the listed commercial products did not clearly show a deployed feature that explains diagnostic conclusions in clinician-facing causal language. Siemens has published research in this direction, but we treat that as a research precedent rather than a clearly deployed commercial feature: `https://arxiv.org/html/2306.10448`.

[42] Lakhani and Sundaram, `https://doi.org/10.1148/radiol.2017162326`.

[43] Pasa et al., `https://doi.org/10.1038/s41598-019-42557-4`.

The novelty claim is deliberately narrow: not that no product has any text output, but that we did not find clear evidence in the reviewed official materials of a deployed feature that explains diagnostic conclusions in clinician-facing causal language. Siemens has published research directly in this direction, which makes the feature technically credible, but that prior work appears to remain in the research-to-product gap.[44]

2. **Input QA module.** Detect poor positioning, clipped lungs, rotation, poor exposure, blur, and non-standard inputs such as photographed films. If quality is inadequate, the system warns, recommends reacquisition, or routes the case to manual review. This is justified by the known effect of acquisition quality and domain shift on chest X-ray AI reliability.[45][46]

3. **Uncertainty / deferral module.** Present "uncertain" as an explicit output state for near-threshold, low-quality, or out-of-distribution cases. This turns thresholding into a safety mechanism rather than the headline contribution. In the reviewed official product materials, we did not find comparably clear evidence of an explicit user-facing deferral workflow. The usability literature supports the need to communicate model confidence carefully.[47]

4. **Structured evidence interface.** Show score, threshold band, quality status, and localized visual evidence in a compact panel.

5. **Post-deployment monitoring module.** Track rates of poor-quality inputs, uncertain cases, deferrals, threshold-borderline cases, reviewer overrides, and possible drift in score distributions over time. Reviews and vendor messaging both support the need for post-launch monitoring, but this is treated here as a supporting operational layer rather than the clearest novelty claim.[48][49]

4.3 Why This Is the Best Fit

- **It is cohesive.** The feature set is not a grab bag if framed around one pipeline: *check the input, qualify the prediction, and monitor the system after deployment.*

- **It is implementable this month.** A quality gate, uncertainty rule, and monitoring interface are much more feasible than building a clinically faithful multimodal explanation engine.

- **It is evidence-backed.** The literature repeatedly identifies real-world workflow integration, local evaluation, and post-launch monitoring as unresolved problems in chest radiography AI.[50][51]

- **Why not make it only a monitoring paper?** A monitoring-only contribution is feasible, but too thin on its own because post-deployment metrics need upstream events to observe. In practice, the most meaningful things to monitor are exactly the outputs of the QA and uncertainty layers, so a broader workflow-layer framing is stronger and better integrated.

[44] Siemens research precedent: `https://arxiv.org/html/2306.10448`.
[45] Mongan et al., `https://pmc.ncbi.nlm.nih.gov/articles/PMC10128969/`.
[46] Generalization and shortcut-learning concerns: `https://pmc.ncbi.nlm.nih.gov/articles/PMC10047562/`.
[47] Oakden-Rayner et al., `https://pubmed.ncbi.nlm.nih.gov/36189838/`.
[48] Mongan et al., `https://pmc.ncbi.nlm.nih.gov/articles/PMC10128969/`.
[49] Lunit INSIGHT CXR / monitoring positioning, `https://www.lunit.io/en/ai-radiology-software/insight-cxr/`.
[50] Mongan et al., `https://pmc.ncbi.nlm.nih.gov/articles/PMC10128969/`.
[51] Systematic review, `https://pmc.ncbi.nlm.nih.gov/articles/PMC9955112/`.

5 Paper Outline

5.1 Abstract Outline

1. **Background:** TB remains a high-burden problem in the Philippines, and CAD for chest radiography is already sufficiently mature to motivate real deployment concerns rather than purely algorithmic ones. WHO has recommended CAD for TB screening since 2021, and six products met WHO performance standards in the 2025 update.[52][53]

2. **Problem:** Existing literature and products show strong detection capability, but workflow fit, local validation, quality assurance, and reliability signaling remain underdeveloped in routine hospital use.[54][55]

3. **Approach:** Present a reliability layer composed of input QA, uncertainty-aware deferral, a structured evidence panel, and monitoring/audit functions around a TB CAD model.

4. **Expected contribution:** A practical hospital-facing system design and evaluation framework for safer and more teachable TB CAD deployment.

5.2 1. Introduction and Evidence Gap

- Introduce Philippine TB burden and the appeal of chest X-ray triage.
- Establish for a non-specialist reader that CAD feasibility is already strong: WHO has recommended CAD for TB screening since 2021, and six products met WHO performance standards in the June 2025 update.[56][57]
- Summarize the broader chest X-ray AI evidence base: systematic reviews report strong overall performance, and newer reader studies show that AI assistance can elevate non-radiologists to radiologist-level performance on chest X-ray tasks.[58][59]
- Add a concrete clinical acceptance signal: in a real-world observational study, radiologists showed complete agreement with model findings in 86.5% of cases, suggesting that the field is beyond the stage of asking only whether AI can reach useful accuracy.[60]
- Introduce the workflow evidence gap directly inside the introduction: reviews report limited prospective workflow studies, limited monitoring discussion, and incomplete understanding of clinician interaction with AI outputs. Usability testing also found that many users believed

[52] WHO policy statement overview, https://www.who.int/publications/i/item/9789240110373.
[53] WHO TB CAD approval update, https://www.who.int/news/item/11-06-2025-who-approves-six-software-products-for-co
[54] Mongan et al., https://pmc.ncbi.nlm.nih.gov/articles/PMC10128969/.
[55] Systematic review, https://pmc.ncbi.nlm.nih.gov/articles/PMC9955112/.
[56] WHO policy statement overview, https://www.who.int/publications/i/item/9789240110373.
[57] WHO TB CAD approval update, https://www.who.int/news/item/11-06-2025-who-approves-six-software-products-for-co
[58] Systematic review, https://pmc.ncbi.nlm.nih.gov/articles/PMC9955112/.
[59] Anderson et al., https://www.nature.com/articles/s41598-024-76608-2.
[60] Real-world observational study, https://pubmed.ncbi.nlm.nih.gov/34930738/.

chest X-ray AI tools would be more useful in understaffed radiology departments or for trainees and non-radiologist specialties.[61][62][63]

- Transition from "accuracy solved enough" to the actual paper gap: unreliable image inputs, uncertainty communication, non-radiologist use, local workflow fit, and limited post-launch monitoring.
- End with the thesis that the paper focuses on a reliability-centered deployment layer for hospital TB CAD.

5.3 2. System Framing and Design Goals

Prototype-oriented research question: Can an academic prototype that integrates clinician-facing multimodal explanation, input QA, uncertainty-aware deferral, and monitoring on top of a mature TB CAD backbone provide a more implementation-ready hospital screening workflow than a classifier-only baseline under preliminary evaluation?

Design goals:

1. Reject or flag unsafe inputs before inference.
2. Avoid overconfident automation by explicitly deferring uncertain cases.
3. Provide clinically grounded explanations of flagged findings in natural language.
4. Present evidence in a compact workflow-friendly interface.
5. Monitor deployment quality over time.

Features considered but excluded: A dedicated trainee or educational mode was considered but excluded from the prototype. A targeted search for primary evidence of end-user demand (surveys, focus groups, or co-design workshops in which residents, clinical officers, or radiographers explicitly requested a dedicated trainee mode in a CXR AI tool) did not surface clear findings; signals that do exist call for external AI curriculum rather than an in-tool trainee feature. Separately, a study of residents reading chest radiographs with AI as a second reader found sensitivity and accuracy gains that disappeared once AI was removed, suggesting dependence rather than durable skill transfer,[64] so a naively implemented trainee mode could reinforce the same failure pattern. Trainee-oriented functionality is therefore folded into the multimodal explanation module, which already produces natural-language descriptions accessible to less-experienced readers. A dedicated trainee mode is retained as future work pending primary user research and a study design that measures learning rather than assisted performance. Full autonomous report generation, formal cost-effectiveness modeling, and smartphone-captured analog radiograph support are similarly excluded and listed as future directions.

[61] Mongan et al., `https://pmc.ncbi.nlm.nih.gov/articles/PMC10128969/`.

[62] Systematic review, `https://pmc.ncbi.nlm.nih.gov/articles/PMC9955112/`.

[63] Cheung et al., `https://journals.sagepub.com/doi/10.1177/08465371221131200`.

[64] Chassagnon et al., "Learning from the machine: AI assistance is not an effective learning tool for resident education in chest X-ray interpretation," *European Radiology*, 2023, `https://pubmed.ncbi.nlm.nih.gov/37572190/`.

5.4 3. System Architecture

- Describe data flow from acquisition to QA gate, classifier, uncertainty logic, multimodal explanation layer, UI, and dashboard.
- Make explicit that the explanation layer is the lead prototype addition, with Siemens' prior research serving as technical precedent rather than evidence of an already deployed commercial feature.
- Clarify where thresholding appears: as a configurable component inside reliability logic, not the main novelty.
- Include a note that photographed-film detection can be treated as a special case under QA and deferral.

5.5 4. Reliability Layer Components

5.1 Input quality gate

- Define quality labels and actions: acceptable, borderline, repeat recommended, manual review required.
- Explain relevance to low-resource and non-standard acquisition settings.

5.2 Uncertainty-aware deferral

- Define uncertain outputs from threshold proximity, poor quality, or OOD cues.
- Explain why deferral is safer than forced binary prediction.

5.3 Structured evidence panel

- Show TB score, risk band, image-quality warning, and localized evidence.
- Connect this panel to both visual and textual explanation outputs.

5.4 Clinically grounded multimodal explainability

- Explain the planned VQA / natural-language explanation layer and the kinds of outputs it should generate for clinicians, emphasizing clinician-facing reasoning such as linking a suspected TB conclusion to visually localized findings.
- Position Siemens' published research as evidence that this feature direction is technically grounded, while noting that the reviewed official commercial product materials do not yet clearly show this as a deployed radiologist-facing feature.[65]

[65] Siemens research precedent: `https://arxiv.org/html/2306.10448`.

- State explicitly that this is presented as a prototype feature implementation, not as a fully validated clinical explanation system.

5.5 Post-deployment monitoring

- Specify metrics: poor-quality input rate, uncertain-case rate, deferral rate, borderline-threshold rate, override rate, repeat-image rate, score drift, and subgroup breakdowns if available.
- Argue that this is more TB hospital-specific than generic enterprise analytics, and note that this monitoring layer could be described as the narrowest standalone contribution if the project scope needs to be reduced further.

5.6 5. Preliminary Evaluation Plan

- **Technical:** classifier performance, quality gate accuracy, uncertainty behavior, explanation examples, and failure-mode examples.
- **Prototype workflow evaluation:** simulated hospital cases, clinician or student walkthroughs, qualitative usability feedback, and timing measures if feasible.
- **Reliability outcomes:** proportion of cases filtered by QA, proportion deferred, override behavior, and what failure modes are caught by the safeguards.
- **Claim boundary:** present the system as an academic prototype with preliminary evaluation, not as a clinically validated workflow intervention; in particular, explainability results should be framed as implementation and example outputs rather than as proof of clinical reasoning validity.

5.7 6. Discussion

- Emphasize that the paper's value is not a new backbone but a richer prototype implementation layer around mature TB CAD, with multimodal explanation as the lead feature.
- Explain why thresholding alone was not selected as the main contribution, even though local thresholds remain important.[66][67]
- Explain why cost-effectiveness and smartphone-film workflows are future directions unless supported by stronger local data.
- Reiterate why this narrative fits a practical medical deployment paper.

[66] WHO calibration guidance, https://wkc.who.int/resources/publications/i/item/determining-the-local-calibration-of-computer-assisted-detection-(cad)-thresholds-and-other-parameters.

[67] Philippine qXR study, https://bmcglobalpublichealth.biomedcentral.com/articles/10.1186/s44263-025-00198-y.

5.8 7. Conclusion

Summarize the paper as a shift from "can TB CAD classify chest X-rays?" to "how can an academic prototype integrate reliability and clinician-facing explanation around mature TB CAD for hospital workflow use?"